Self Invoke Function

A self-invoking function—more commonly known as an Immediately Invoked Function Expression (IIFE)—is a JavaScript function that runs automatically as soon as it is defined Immediately Invoked Function Expression (IIFE)—is a JavaScript function that runs automatically as soon as it is defined. Unlike normal functions, you do not need to explicitly call it later in your code.

Standard Syntax

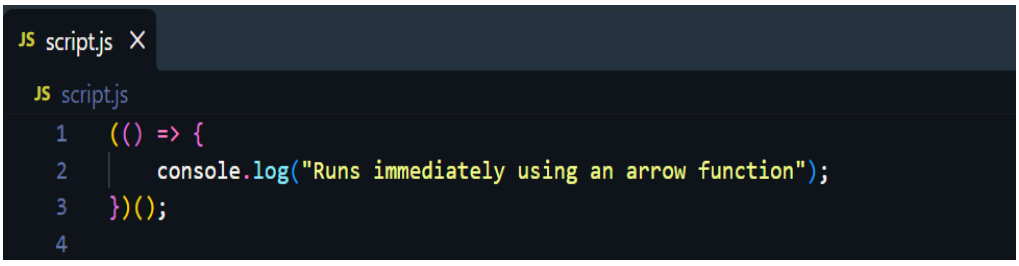
To create an IIFE, you wrap an anonymous function inside a grouping operator () to turn it into an expression, and then append a second pair of () to execute it immediately:



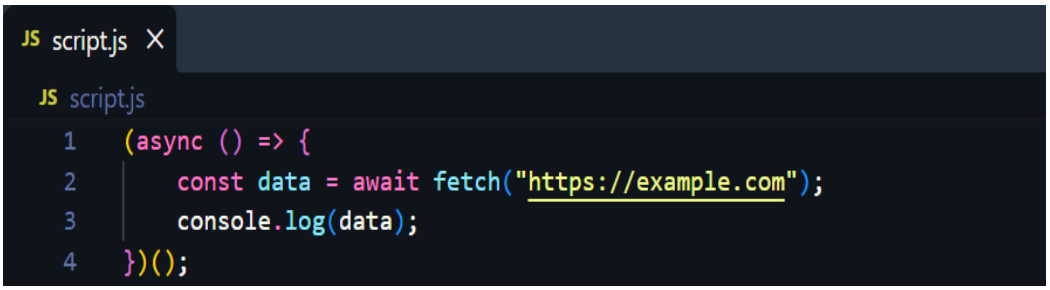
Modern Syntax Variants

You can write self-invoking functions using modern JavaScript conventions, such as arrow syntax or async execution:

Arrow Function Variant:



Async IIFE (useful for await at the top level):



Why Use a Self-Invoking Function?

1. **Avoid Global Scope Pollution:** Variables declared inside the function stay inside the function. This prevents you from accidentally overwriting variables elsewhere in your script.

2. **Initialization Code:** They are perfect for code that only needs to run once, such as setting up event listeners, configuring plugin options, or calculating an initial state on page load.

3. **Creating Data Privacy (Closures):** You can use an IIFE to hide variables while exposing a public method to interact with them.

Example: Data Privacy with Closures

In this example from the freeCodeCamp Forum, the counter variable is completely protected from the outside world, yet it maintains its state through the inner function:

